

DUSTBADGE · CALCULATION

DustBadge sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	Worn version in the mass estimate (rear shell 1 mm deeper, foam pad): 120.7 g against 122 g; cost USD 91.50, USD 0.50 over the target, so R15 is not met; low-voltage cutoff threshold added (section A)
0.6	2026-10-02	Amish Chadha	R9 target 122 g (decided by Amish, 2026-10-02) in the requirement table
0.5	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.4	2026-09-30	Amish Chadha	Design made constructable (DBG-DDR-003): shell screws, gasket, port seals, ribs and clip screws added; mass 119.9 g, margin 0.1 g; drop retention paths stated
0.3	2026-09-26	Amish Chadha	Budget approved by Amish; budget \$91 covers the priced BOM, so R15 is met (DBG-DDR-002 v0.2)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002): 2,000 mAh cell, R2 restated, R3 reference rule, R11 sun use rule; results table updated

0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (power and run time, charging, badge temperature, PM4 against the respirable convention, range, error budget, alert logic, log, breathing zone, mass, drop, cost)
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DustBadge sizing calculations

On paper, DustBadge meets ten of its fifteen requirements (six by calculation, four by design), has two at risk, cannot verify one at TRL 3 and misses two (R14 and, by USD 0.50 against the value-engineering target, R15). This version applies the recommendations Amish accepted on 2026-09-25 (DBG-DDR-002). The 2,000 mAh cell (was 1,500 mAh) runs 12.8 h at the sensor's maximum current and 0 °C (was 9.6 h), so R7 moves from at risk to met, at the cost of 8 g and \$3. The design for construction (DBG-DDR-003, v0.4 of this note) adds the shell and clip screws, the gasket, the port seals and the locating ribs, which brought the badge to 119.9 g against 120 g and the estimated parts cost to \$91.00, \$1 over the \$90 value-engineering target; on 2026-09-26 Amish set the target at \$91 to match the priced BOM (DBG-DDR-002). The worn version decided on 2026-10-02 (v0.7 of this note) deepens the rear shell by 1 mm and adds a foam pad behind the cell: the badge is 120.7 g against the relaxed limit of 122 g, so R9 stays met, but the parts cost is \$91.50, \$0.50 over the \$91 target, so R15 is no longer met. The firmware low-voltage cutoff is set at 3.30 V (section A). R2 is restated to the sensor's 0 to 1 mg/m³ range with over-range minutes flagged and counted, and is met on paper. R3 keeps $\pm 25\%$ with a 4.2 L/min, two-shift reference at low-dust, high-silica sites: ± 21 to $\pm 22\%$ with a factor per task on the same badge, but ± 29 to $\pm 30\%$ with one factor shared across badges, so it is at risk. R11 now carries the use rule to wear the badge shaded above 40 °C in full sun, which keeps the shell at or below 57.8 °C, under the sensor's 60 °C limit; it is at risk because the rule depends on the wearer and the sensor is outside its best-performance range in sun above about 22 °C. Intrinsic safety (R14) remains not met and out of scope. The TRL 3 calculations had already added a humidity and temperature sensor and a thermistor charger to the TRL 2 concept. Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [A3], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. They do not show that a worker's air is safe, and they are not a substitute for co-located filter sampling, chamber checks or electrical safety checks on the lithium cell. See DBG-PRC-001, Safety.

■ Scope and method

The note checks every requirement in DBG-REQ-001 v0.4 against the design in DBG-PRC-001 v0.4 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, derived dimensions and mass breakdown, so the envelope, shell volumes, inlet position and cell space used here are the ones in the STEP files and in drawing DBG-DWG-001. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py`.

The design case is a worker in a surface quarry or a stone fabrication shop on a 12 h shift, badge worn at the collar or on an upper harness strap, ambient 0 to 45 °C, 10 to 90 % RH, US OSHA and MSHA limits by default (decision D3 in DBG-DDR-001, decided by Amish on 2026-09-25 in DBG-DDR-002).

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Particle sensor	45 to 55 mA typical and 65 mA maximum at 4.5 to 5.5 V; 0 to 1,000 µg/m ³ ; PM4 precision ±25 µg/m ³ below 100 µg/m ³ and ±25 % from 100 to 1,000 µg/m ³ ; drift 1.25 µg/m ³ or 1.25 % per year; best performance at 10 to 40 °C and 20 to 80 % RH; operating -10 to 60 °C and 0 to 95 % RH; new readings every second; 26.3 g	Sensirion SPS30 datasheet, v2.0, June 2023
Electronics	5 V boost 85 % efficient; controller, BLE and humidity sensor 3 mA from the cell; vibration motor 80 mA at 3 V, 20 alerts of 2 s per shift	Typical module figures; to confirm by measurement at a later TRL
Cell	2,000 mAh, 3.7 V, 11.5 mm thick, 38 g (DBG-DDR-002 D8; was 1,500 mAh, 30 g); 80 % usable; 85 % of capacity at 0 °C; charged at 500 mA; protection cut-out assumed at 2.75 V	Typical LiPo data; the protection value is to be taken from the bought cell's datasheet
Badge temperature	Combined surface coefficient 10 W/(m ² ·K); solar absorptance of yellow PETG 0.45; 1,000 W/m ² on the front face; rear face against the wearer treated as adiabatic	Screening values (assumed)
Reference sample	Cyclone at 2.5 L/min for 8 h; at low-dust, high-silica sites a 4.2 L/min cyclone over two shifts (DBG-DDR-002 D9); weighing and blank uncertainty 0.010 mg per filter; gravimetric method ±10 %	Assumed typical values
Error terms	Site factor change between days and tasks ±10 % (factor per task) or ±20 % (per site); position and screen ±10 %; humidity below 80 % RH ±5 %; non-linearity after the site factor ±10 %; badge-to-badge after a CalRig check ±10 %	Assumed; each to be measured
Silica fractions	10 % for a typical quarry, 80 % for quartz-rich engineered stone	Assumed for illustration; a site filter sample sets the real value

■ A. Power and run time (R7)

The 2,000 mAh cell holds 7.40 Wh, of which 5.92 Wh is usable, and 5.03 Wh at 0 °C [A1]. The motor adds only 2.7 mWh per shift [A2]. The sensor's fan and laser dominate.

Table 2. Run time by sensor current [A3].

Sensor current	Power from the cell	Run time at 25 °C	Run time at 0 °C
45 mA (typical, low)	276 mW	21.5 h	18.2 h
55 mA (typical, high)	335 mW	17.7 h	15.0 h
65 mA (maximum)	393 mW	15.0 h	12.8 h

Covering 12 h at the maximum current and 0 °C needs about 1,880 mAh [A4], or about 1,600 mAh at 25 °C [A5]. The 2,000 mAh cell, 11.5 mm thick, leaves 1.5 mm to the rear wall in the worn version (0.5 mm in the bench version), of which the 1 mm foam pad takes 1.0 mm, and a 122 mAh (7 %) margin over the worst case [A6]. With the 1,500 mAh cell of v0.1 the run time was 13.3 h typical and 9.6 h in the worst case, and R7 was at risk. R7 is now met on paper.

The firmware stops the cell short of its own protection cut-out (decision of 2026-10-02, DBG-DDR-003 A2): the controller switches the boost off through its enable input and sleeps when the cell reaches 3.30 V under load, which is 0.55 V above the assumed protection cut-out of 2.75 V, and it restarts the boost only above 3.50 V or on USB power [A7]. The cutoff does not change the 80 % usable fraction behind the run times above; that is an assumption to confirm against the bought cell's discharge curve. The threshold is the figure a future firmware sketch must use; no firmware exists at TRL 3.

■ B. Charging

At 500 mA (0.25 C) the 2,000 mAh cell charges in about 4.2 h, 3.2 h at constant current plus about 1 h of taper [B1] (3.4 h for the 1,500 mAh cell), so a badge charged after a 12 h shift is still ready for the next. The charger module now takes the cell's thermistor so that charging stops outside 0 to 45 °C; at TRL 2 this rule was written but nothing enforced it.

■ C. Badge temperature (R11)

In shade the electronics warm the badge by 3.3 K. In full sun on the front face the rise is 17.8 K: the shell reaches about 48 °C at 30 °C ambient and about 63 °C at 45 °C ambient [C1]. The sensor's 60 °C operating limit is therefore passed in full sun above about 42 °C ambient, and its 40 °C best-performance limit above about 22 °C ambient [C2]. The sensor draws outside air through its own path, so the air it measures stays near ambient, but its electronics and the cell sit at shell temperature. Without a rule, R11 is not met in full sun at 45 °C. Amish accepted the use rule on 2026-09-25 (DBG-DDR-002 D10): wear the badge shaded, under a vest flap or out of direct sun, when the ambient temperature is above 40 °C in full sun. The worst case is then 57.8 °C in full sun at 40 °C, or 48.3 °C shaded at 45 °C, both under the 60 °C operating limit with a 2.2 K margin [C3]. R11 is at risk: the rule depends on the wearer, the margin is small and the screening coefficients are assumed, and in sun above about 22 °C the sensor's electronics are above their 40 °C best-performance limit. A lighter shell tested under a solar lamp is TRL 4 work and is on hold.

■ D. PM4 against the respirable convention (R1)

The ISO 7708 respirable convention passes half of particles at 4.0 µm aerodynamic diameter [D1]. For quartz (2.65 g/cm³), geometric and aerodynamic diameters differ by a factor of about 1.6: a 2.5 µm grain is 4.1 µm aerodynamic and about half respirable, and a 4 µm grain is 6.5 µm aerodynamic and only 12 % respirable [D2]. An optical PM4 channel would therefore over-include coarse quartz if it sized it correctly. In practice the SPS30 calculates PM4 from the distribution of the particles it detects best, which are fine, so it tends to under-see coarse grains. The two errors act in opposite directions and neither is known for site dust [D3]. PM4 remains the closest available output, so R1 is met by design as a proxy, and the site factor must absorb the difference.

■ E. Working range (R2)

The sensor covers 0 to 1,000 µg/m³. R2 is restated (DBG-DDR-002 D11) to this range with over-range minutes flagged and counted, for the first sectors in D6; the TRL 2 target of 5 mg/m³ was only 20 % covered [E1]. What matters for the alert is where the silica limits fall in dust terms.

Table 3. Respirable dust at the RCS action level and limit, by silica fraction [E2].

Silica fraction	Dust at the 25 µg/m³ action level	Dust at the 50 µg/m³ limit	Limit in range
2.5 %	1,000 µg/m³	2,000 µg/m³	No
5 %	500 µg/m³	1,000 µg/m³	Yes, at the edge
10 %	250 µg/m³	500 µg/m³	Yes
30 %	83 µg/m³	167 µg/m³	Yes
80 %	31 µg/m³	62 µg/m³	Yes

For dust with 5 % silica or more the limit lies inside the range, and for 2.5 % or more the action level does [E3]. Above the range, readings are not trustworthy, so the firmware logs an over-range minute at the range limit, flags it and counts it toward the projection at no less than that limit, which makes the badge alert rather than go quiet [E4]. R2 as restated is met on paper. Peaks in quarries and artisanal mines can be far above 1 mg/m³, and the badge will report only that they were over range; for dust below 5 % silica the limit lies above the range, so the badge can say only that the dust was over range, not by how much.

■ F. Error budget for the shift average (R3)

A site factor from a co-located filter sample is what turns the optical reading into a respirable dust value. The filter sample itself limits the result at low concentrations: 2.5 L/min for 8 h draws 1.20 m³ of air [F1], so at the action level for quartz-rich stone (31 µg/m³ of dust) the filter collects only 0.037 mg, and a 0.010 mg weighing uncertainty is already 27 %.

Table 4. Combined uncertainty of the shift average at the action level (root sum of squares) [F2, F3].

Case	Dust	Filter mass	Same badge, factor per task	Factor shared across badges, per site
Quartz-rich stone, 80 % silica	31 µg/m³	0.037 mg	±34 %	±39 %
Quarry, 10 % silica	250 µg/m³	0.300 mg	±21 %	±29 %
Quartz-rich stone, 4.2 L/min cyclone over 2 shifts (D9 rule)	31 µg/m³	0.126 mg	±22 %	±30 %

Amish accepted on 2026-09-25 keeping the ±25 % target and requiring a 4.2 L/min cyclone over two shifts at low-dust, high-silica sites (DBG-DDR-002 D9). With that rule the target is met on paper in both cases when the reference is taken on the same badge for each task (±21 % quarry, ±22 % quartz-rich stone), and missed when one factor is shared across badges per site (±29 %, ±30 %) [F2, F3]. Without any site factor the PM4 precision alone is ±25 µg/m³, 80 % of the dust level at the action level for quartz-rich stone [F4]. R3 is at risk. Every term except the sensor's own figures is assumed and must be measured.

■ G. Silica estimate and alert logic (R4, R5, R6)

The sensor gives new readings every second; the firmware logs 1 min means, multiplies calibrated PM4 by the site silica fraction to give the RCS estimate, and projects the 8 h TWA as the dose so far plus the last 60 min mean rate for the rest of the shift. The projection starts once 60 min of data exist; before that the badge alerts only on the dose so far and on peaks. A brief peak alert fires when a 1 min mean exceeds 250 $\mu\text{g}/\text{m}^3$ RCS, ten times the action level [G2]. Five scenarios on an 8 h shift test the logic [G1].

Table 5. Alert behavior in five scenarios (RCS, $\mu\text{g}/\text{m}^3$) [G1].

Scenario	8 h TWA	Action-level alert	Limit alert	Peak alerts	Correct?
S1 steady 30 all shift	30.0	Minute 60	None	0	Yes
S2 background 10, 2 min dry cut at 1,000 each hour	43.0	Minute 60	None	8	Yes
S3 background 15, one 1 min truck pass at 110	15.2	None	None	0	Yes, no nuisance alert
S4 background 5	5.0	None	None	0	Yes
S5 background 10, one 2 min dry cut at 1,000	14.1	Minute 60	None	1	Early warning; the shift ends below the action level

The logic warns before the limit in every case where the shift ends above the action level, and it ignores a passing truck. S5 shows the cost of projecting: a single heavy cut in the first hour projects over the action level although the shift ends at 14 $\mu\text{g}/\text{m}^3$. That is acceptable for a warning, since the same cut repeated each hour gives S2, but workers should be told what the alert means. R4, R5 and R6 are met by design; R6 is also met on paper for these scenarios.

■ H. Shift log (R12)

At 16 bytes per minute a 12 h shift takes 11.5 kB, and 30 shifts take 346 kB, 16 % of the module's 2 MB flash, so the badge holds about 182 shifts [H1]. At a conservative 2 kB/s a shift transfers to a phone in about 6 s [H2]. R12 is met on paper.

■ I. Breathing zone (R8)

With the badge worn 210 mm below and 70 mm to the side of the nose and mouth, the inlet is 242 mm from them [I1]. The badge center may sit up to 269 mm below the nose and mouth before the 300 mm limit is passed [I2], so a collar or upper-strap position meets R8 and a low shirt pocket may not. R8 is met on paper for the stated mount.

■ J. Size and mass (R9)

The printed shells weigh 16.5 g (front) and 16.0 g (rear) from the model volumes. The rear shell was 15.4 g in the bench version; the worn version (decided on 2026-10-02, DBG-DDR-003 A1) deepens it by 1 mm and lengthens its tubes, bosses and locating ribs to suit. The foam pad behind the cell adds 0.2 g. The gasket and the two port seals, from the model volume in TPU, weigh 0.8 g. The sensor (26.3 g) and the 2,000 mAh cell (38 g) are more than half the badge [J1]. The total is 120.7 g (119.9 g in the bench version), 1.3 g under the relaxed 122 g limit and 0.7 g over

the former 120 g; the worn version adds 0.8 g, not the 1.5 g first estimated in DBG-DDR-003. The envelope is 64 x 52 x 34 mm including the clip against 75 x 55 x 35 mm [J2, J3]; the light pipe stands 1 mm proud of the front face, so the greatest depth is 35.0 mm, exactly the limit and the same 1 mm that was there in the bench version. R9 is met on paper against 122 g.

■ K. Drop and ingress (R10)

A 1.5 m drop reaches 5.42 m/s with 1.78 J. If the shell corner crushes 1 to 2 mm, the mean deceleration is 750 to 1,500 g, so the sensor needs 194 to 387 N of retention and the heavier cell 280 to 559 N (was 221 to 441 N) [K1]. The model now gives each a load path (DBG-DDR-003): the sensor sits on its two port seals, under two stop ribs in the front shell, with the carrier board behind it; the board is clamped between the three front bosses and three tubes in the rear shell by M2 screws; the cell sits between the board and the rear shell, located by four ribs; whether PETG bosses and a foam-backed cell survive these loads, and whether the gasketed joint reaches IP54 for the electronics, cannot be shown without a test. R10 is not verifiable at TRL 3.

■ L. Cost (R15)

The BOM has 13 lines totaling an estimated \$91.50 against the value-engineering target of \$91 (budget_usd , a hypothetical control target, not a limit), \$0.50 over it; the particle sensor is 52 % of the cost [L1]. Value-engineering target: USD 91. Estimated cost of the constructable design: USD 91.50 (USD 0.50 over the target) [L2]. The \$0.50 is the foam pad (line 13) added for the worn version on 2026-10-02; the extra PETG in the deeper rear shell is about a cent and is inside line 9. The 2,000 mAh cell added \$3 earlier (was \$88.00), and Amish moved the target to \$91 on 2026-09-26 to match the priced BOM (DBG-DDR-002, O3). R15 is not met on paper: raising the target to \$92 is proposed to Amish in the design decisions register. The optional breakaway lanyard decided for TRL 4 (about \$1) is not in this estimate.

■ M. Results against every requirement

Table 6. Requirement status from this note (v0.7).

ID	Requirement	Value	Target	Status
R14	Safe in hazardous atmospheres	Not intrinsically safe	Certified intrinsically safe	Not met , out of scope
R3	Accuracy after site calibration	±21 % (quarry), ±22 % (quartz-rich stone, D9 reference) with a factor per task; ±29 to ±30 % with a shared factor	±25 % of a gravimetric shift average; D9 reference at low-dust, high-silica sites	At risk (was not met; assumed terms)
R11	Work in site conditions	Shell at most 57.8 °C under the use rule; best performance only to 40 °C and 80 % RH; humidity flag	0 to 45 °C, 10 to 90 % RH, readings flagged; worn shaded above 40 °C in full sun	At risk (was not met)
R10	Survive the site	750 to 1,500 g on a 1.5 m drop; IP54 by gasket, untested	IP54 electronics; 1.5 m drop	Not verifiable at TRL 3
R2	Cover the working range	0 to 1,000 µg/m³; over-range flagged and counted	0 to 1 mg/m³ with over-range flag (restated)	Met on paper (was not met against 5 mg/m³)
R6	Warn before a limit is reached	Five scenarios behave as intended; one early warning (S5)	Projected TWA alerts at 25 and 50 µg/m³	Met on paper
R7	Last a full shift	17.7 h typical at 25 °C; 12.8 h at maximum current and 0 °C	12 h continuous	Met on paper (was at risk)
R8	Wearable in the breathing zone	Inlet 242 mm from nose and mouth	300 mm or less	Met on paper
R9	Light and small	120.7 g with the worn version; 64 x 52 x 34 mm	122 g (relaxed from 120 g by Amish, 2026-10-02); 75 x 55 x 35 mm	Met on paper, 1.3 g margin
R12	Keep a shift log	11.5 kB per shift; 182 shifts; 6 s transfer	30 shifts; BLE export	Met on paper
R15	Low cost and buildable	\$91.50; perfboard carrier	\$91; no custom PCB	Not met by \$0.50 (foam pad added 2026-10-02; was within the target at \$91.00)
R1	Measure the respirable fraction	PM4 as proxy; size cut offset in both directions (section D)	PM4 as ISO 7708 proxy	Met by design
R4	Estimate silica exposure	Calibrated PM4 x site fraction, labeled	As stated	Met by design
R5	Update often enough	1 s readings, 1 min log, TWA every minute	As stated	Met by design
R13	Protect the worker's data	Badge and phone only; sharing by the worker	As stated	Met by design

Counts: 2 not met (R14, R15), 2 at risk, 1 not verifiable at TRL 3, 6 met on paper, 4 met by design (v0.6: 1 not met, 2 at risk, 1 not verifiable, 7 met on paper, 4 met by design; v0.1: 4 not met, 1 at risk, 1 not verifiable, 5 met on paper, 4 met by design).

■ Checks against the TRL 2 figures

Table 7. TRL 2 claims checked (as issued in v0.1, with the 1,500 mAh cell; the v0.2 values are in Table 6).

TRL 2 claim (DBG-PRC-001 v0.2)	This note	Action
About 335 mW from the cell	335 mW at 55 mA; 393 mW at 65 mA	Stands
About 13 h typical, about 11 h worst case	13.3 h and 11.3 h; also 11.3 h at 0 °C	Precis updated with the cold case
Charge about 3.5 h	3.4 h	Stands
Sensor precision $\pm 5 \mu\text{g}/\text{m}^3$ plus 5 % at the action level for quartz-rich stone	That is the PM2.5 figure; PM4 is $\pm 25 \mu\text{g}/\text{m}^3$ below 100 $\mu\text{g}/\text{m}^3$	Precis corrected
Log about 12 kB per shift, 30 shifts about 350 kB	11.5 kB and 346 kB	Stands
About 220 mm from badge to nose	Inlet 242 mm from the nose and mouth midpoint	Precis and requirements updated
About 110 g	111.8 g	Precis updated
64 x 52 x 30 mm	64 x 52 x 30 mm shells, 33 mm with the clip	Precis updated
About \$85	\$88.00 with the humidity sensor and thermistor charger	Precis, BOM notes and README updated
Badge temperature not estimated	About 63 °C shell in full sun at 45 °C	R11 status changed to not met